CASE STUDY REPORT

E-Waste, Recycling and Circular Economy: Ten Case Studies

Japan · Switzerland · India's E-Waste Rules 2022 · EPR · Bengaluru · Mumbai · Kerala · Pune SWaCH · Dharavi · Dell

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Each case follows the required format: Executive summary, Background, Strategy/initiative, Impact and results (data table), SWOT analysis, Comparison with India, Recommendations.

CASE STUDY 1 OF 10

1. Why Japan Recycles More Than Most Countries: Lessons for India

1. Executive Summary

Japan has little land, little landfill space and almost no domestic mineral resources, so it built its recycling system on three pillars: product-specific laws that divide responsibility between citizens, municipalities and producers; strict household sorting; and industry-led technology and eco-design. The results are striking for some streams (about 87% of PET bottles were recycled in FY2022) but modest overall: only about 20% of Japan's total waste is recycled and over 70% of municipal waste is incinerated. The lesson for India is not to copy Japan wholesale, but to borrow its clarity of roles, source segregation and design-for-recycling discipline.

2. Background

Japan's approach rests on the idea of a 'sound material-cycle society'. From the mid-1990s it passed a family of recycling laws: the **Containers and Packaging Recycling Law** (1995; PET bottles were brought under it in 1997), the **Specified Home Appliance Recycling Law** (enacted 1998, in force April 2001, covering air conditioners, TVs, refrigerators/freezers and washing machines/dryers), the Green Purchasing Law (in force April 2001) and later a law for small household appliances (2013).

Before the packaging law, municipal PET collection was in single digits. By FY2003 the national PET collection rate had reached 61%, against roughly 20% in the US and 30% in Europe, and by 2019 almost every municipality (99.6%) collected PET bottles separately.

3. Strategy / Initiative

- **Shared responsibility by law:** consumers sort waste (and pay a recycling fee for regulated appliances); municipalities collect the separated materials; producers, retailers and businesses that use packaging fund and carry out recycling.
- **Separate collection and strict sorting:** households sort plastics, paper, PET, glass and metals according to local calendars. Some towns go much further; Kamikatsu, for example, sorts into about 45 categories and reports recycling rates of around 80%.
- **Eco-design rules:** labels on PET bottles are designed to be torn off easily, and coloured PET bottles were phased out because colour lowers the value of recycled flakes and pellets.
- **Industry coordination:** the Council for PET Bottle Recycling and beverage companies track collection and recycling, lighten bottles and push bottle-to-bottle recycling, with an aim of 100% effective use of PET by 2030.
- **Public campaigns:** the Tokyo 2020 Medal Project (April 2017 to March 2019) asked citizens to donate old phones and small electronics; every one of the ~5,000 Olympic and Paralympic medals was made from the recovered metal.

4. Impact & Results

Table 1.1 summarises reported outcomes.

Table 1.1 Japan: recycling indicators

Indicator	Result	Note / source year
PET bottle recycling rate	≈ 87%	FY2022, Council for PET Bottle Recycling
PET collection / recycling rate	≈ 90% / ≈ 85%	IGES-CCET presentation
Municipalities collecting PET separately	99.6%	2019

Indicator	Result	Note / source year
PET collection rate, FY2003	61% (US 19.6%, Europe 30%)	Japan for Sustainability, 2005
Overall waste recycling rate	≈ 20%	Over 70% of municipal waste is incinerated
Tokyo 2020: small e-waste collected	78,985 t via 1,621 municipalities	Apr 2017 to Mar 2019
Tokyo 2020: mobile phones collected	6.21 million	Through NTT docomo shops
Tokyo 2020: metal recovered	32 kg gold, 3,500 kg silver, 2,200 kg bronze	≈ 5,000 medals, 100% recycled metal

Exam tip: for plastics, Japan's high 'recycling' numbers include thermal recovery (burning for energy), which is not material recycling.

5. SWOT Analysis

Strengths

- Clear legal split of roles between citizen, municipality and producer
- High public compliance with sorting
- Eco-design and labelling standards
- Strong industry associations and reliable data

Weaknesses

- Overall recycling only ≈ 20%; incineration dominates
- Thermal recovery counted alongside material recycling for plastics
- Fine-grained sorting is costly and labour-intensive for municipalities
- E-waste collection gaps (a 2016 estimate: only 24–30% of e-waste recycled)

Opportunities

- Bottle-to-bottle and chemical recycling to reduce down-cycling
- Urban mining of metals (the Tokyo 2020 model)
- Exporting technology and know-how to Asia

Threats

- Shrinking landfill capacity
- Loss of export markets: PET exports to China fell from ≈ 316,000 t (2017) to under 21,000 t (2018 projection)
- Ageing population may raise collection costs

6. Comparison with India

Table 1.2 compares Japan with India on the main design features.

Table 1.2 Japan vs India

Parameter	Japan	India
Legal basis	Product-specific recycling laws (packaging 1995, appliances 2001)	EPR under Plastic Waste Rules and E-Waste Rules 2022; SWM Rules 2016 require segregation
Household segregation	Near-universal, calendar-based	Patchy; strong in pockets such as Pune (Case 8)
Who collects	Municipality (formal)	Municipal contractors plus informal waste pickers and kabadiwalas
Informal sector	Negligible	Central to material recovery (Cases 8 and 9)
E-waste	Consumer-paid fee for four regulated appliance types; small-appliance law	≈ 43% of estimated e-waste formally recycled in 2023-24; the rest is informal or unrecycled
Who pays	Consumers and producers	Producers, through EPR certificates

Key takeaway: Japan shows what a formal, rule-bound system can achieve; India has to reach similar outcomes through a hybrid that includes its large informal workforce.

7. Recommendations

1. Scale mandatory source segregation city by city, starting with a published dry-waste collection calendar backed by modest fines and rewards.
2. Set design-for-recycling standards (mono-material PET, easy-peel labels, no coloured bottles) through BIS norms and EPR guidelines.
3. Adopt a Japan-style shared-responsibility rule, but pay waste pickers for the collection service (see Case 8).
4. Run a national 'phones-to-medals' style donation drive around a major public event to build the e-waste return habit.
5. Report material recycling separately from energy recovery, and invest in bottle-to-bottle capacity.

Sources: [1] [2] [3] [4] [5] [6] [7] [8] (see References)

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2. Switzerland's E-Waste Success Story

1. Executive Summary

Switzerland runs one of the world's oldest and most successful e-waste systems. Consumers return old appliances free of charge to any seller or collection point; recycling is pre-funded by a small advance recycling fee (ARF) added to the price of every new product; and industry-run organisations (SENS and Swico) organise collection, audit recyclers and manage the money. About 15 kg of e-waste per person was collected in 2021 (SENS plus Swico), and SENS alone recycled 104,751 tonnes in 2025.

2. Background

Producer-run take-back systems began in Switzerland in the 1990s: SENS started with refrigerators and now covers household and lighting equipment, photovoltaics and more, while Swico Recycling covers IT, office and consumer electronics. The state later gave the arrangement a legal base through the **Ordinance on the Return, Taking Back and Disposal of Electrical and Electronic Equipment (ORDEA)**, in force since 1998.

The ordinance obliges consumers to return old devices, requires retailers, manufacturers and importers to take them back, and requires recyclers to treat them in an environmentally sound way. Firms join the industry systems by their own decision, with the ordinance as a legal backstop.

3. Strategy / Initiative

- **Advance recycling fee (ARF):** producers and importers pay a fee per unit sold. SENS manages the fund, which pays for collection, transport, treatment and administration (a pay-as-you-go model, like a pension scheme). In 2025, 1,557 manufacturers and importers were subject to ARFs.
- **Return anywhere, free:** consumers can hand in a device at any seller (even if not bought there) or at an accredited collection point; SENS reported 634 official collection points (excluding satellites) in 2025.
- **Audited recycling network:** SENS contracts licensed recyclers (17 in 2025) under detailed technical rules for storage, handling and reporting, with at least one expert audit a year.
- **Export control:** exporting defective appliances needs a federal permit, so leakage abroad is rare.
- **Ownership by industry:** because producers run and fund the system, they have a stake in keeping costs and quality under control.

4. Impact & Results

Table 2.1 lists key results reported by SENS and other sources.

Table 2.1 Switzerland: e-waste collection results

Indicator	Result	Note / source year
E-waste recycled in SENS system	92,874 t (10.36 kg/person)	2023
E-waste recycled in SENS system	100,762 t (11.14 kg/person)	2024
E-waste recycled in SENS system	104,751 t (11.48 kg/person)	2025
SENS + Swico combined collection	≈ 15.3 kg per person	2021
SENS + Swico earlier peak	136,200 t; 16.5 kg per person	2014
Refrigerator/freezer collection rate	82% (85% in 2023)	2024; collected ÷ sold
Official collection points / recyclers	634 / 17	2025
Firms paying the ARF	1,557 (+6% on 2024)	2025

The collection rate compares waste collected with new appliances sold; it does not count devices kept at home or passed on for reuse.

5. SWOT Analysis

Strengths	Weaknesses
• Free and convenient return for consumers • Pre-funded, transparent ARF; no public money needed • Audited recyclers and export control • Industry ownership of the system	• Relies on consumer effort; devices hoarded at home are not captured • Collection-rate calculation has known inaccuracies (product mix, item-to-kg conversion) • Depends on high income and a strong compliance culture, so it is hard to transplant
Opportunities	**Threats**
• New categories added (photovoltaics, heat pumps, e-cigarettes, lamps) • More reuse and repair before recycling • Recovery of critical raw materials	• Rapid growth of battery-heavy devices and fire risk • Cross-border online sales that bypass the fee • Falling metal prices raising net recycling cost

6. Comparison with India

Table 2.2 compares Switzerland with India. Per-person figures for India are approximate (author's calculation).

Table 2.2 Switzerland vs India

Parameter	Switzerland	India
Legal base	ORDEA (1998); producer-run SENS and Swico	E-Waste (Management) Rules 2022, in force 1 Apr 2023; EPR targets 60% to 80%
Who pays and how	Visible fee at sale, pooled in one fund	Producers buy EPR certificates from recyclers (reported ₹10–50/kg in 2024-25)
Collection network	634 official points plus return at every seller	Few formal points; public bins only piloted (e.g., 50 in Bengaluru, 2022)
Collected per person	≈ 11.5 kg (SENS); ≈ 15 kg incl. Swico	≈ 1.2 kg generated and ≈ 0.5 kg formally recycled (1.75 Mt and 0.76 Mt, 2023-24)
Informal sector	Practically absent	Handles a large share; over half is informally processed or not recycled
Consumer behaviour	Returns items free of charge	Often sells to scrap dealers for cash

> **Key takeaway:** The Swiss system works because return is free and easy and the money is collected up front. India's system pays producers' money to recyclers, but not to whoever collects the item.

7. Recommendations

1. Make retailer take-back compulsory and visible, with drop boxes at every authorised store.
2. Pay for collection: link EPR money to a per-kg payment at the point of hand-over so formal channels can compete with cash buyers.
3. Create an industry-governed, non-profit clearing body with published fees and audited accounts, alongside private PROs.
4. Audit recyclers annually with mass-balance checks and publish the results.
5. Start with high-value, high-volume items (phones, laptops, TVs, refrigerators) before expanding categories.

Sources: [9] [10] [11] [12] [13] (see References)

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3. India's E-Waste (Management) Rules 2022

1. Executive Summary

The E-Waste (Management) Rules, 2022 (notified on 2 November 2022 and in force from 1 April 2023) replaced the 2016 Rules and turned Extended Producer Responsibility (EPR) into a target-and-certificate system run through a CPCB online portal. Producers must meet rising recycling targets (60% rising to 80% by weight), recyclers earn tradable EPR certificates, and refurbishers, solar panels and 106 product types are brought into the net. The design is strong on paper; results depend on enforcement, verification and pricing.

2. Background

CPCB data show that e-waste from the 21 notified product categories more than doubled from 708,445 t in 2017-18 to 1,601,155 t in 2021-22, while only 527,131 t (32.9%) was collected and processed in 2021-22. The 2016 Rules had introduced Producer Responsibility Organisations (PROs) but relied on state-level authorisations and had weak verification, so the government rewrote the framework.

3. Strategy / Initiative

- **Registration:** manufacturers, producers, refurbishers and recyclers must register on the CPCB portal, replacing state authorisation; no entity may deal with an unregistered one.
- **Wider scope:** Schedule I now lists 106 types of electrical and electronic equipment (EEE) under EPR; solar PV modules, panels and cells are added.
- **Recycling targets:** by weight of EEE sold in year Y minus X, where X is the average product life (Table 3.1).
- **EPR certificates:** recyclers generate certificates from verified end-product recovery (quantity computed from end products using conversion factors); producers buy them online to meet targets.
- **Refurbishers:** refurbishing certificates defer a producer's EPR for the extended life of the refurbished product.
- **Enforcement tools:** environmental compensation for shortfalls, verification and audit, limits on hazardous substances such as lead and mercury, and 100% EPR for importers of used EEE that is not re-exported.

Table 3.1 E-waste recycling targets (Schedule III); to be reviewed after 2028-29

Financial year	2023-24	2024-25	2025-26	2026-27	2027-28	2028-29 on
Target (by weight)	60%	60%	70%	70%	80%	80%

4. Impact & Results

Table 3.2 shows the situation before and after the new rules, using CPCB-based figures.

Table 3.2 India: e-waste generation, processing and capacity

Indicator	Result	Note / source year
E-waste generated (21 categories)	708,445 t	2017-18
E-waste generated / processed	1,346,496 t / 354,291 t (26.3%)	2020-21
E-waste generated / processed	1,601,155 t / 527,131 t (32.9%)	2021-22
E-waste generated / recycled	≈ 1.75 Mt / ≈ 0.76 Mt (≈ 43%)	2023-24, media summary of CPCB data
Registered recyclers	322; capacity ≈ 2.21 Mt/yr	February 2025
Registered refurbishers	72; capacity ≈ 92,042 t/yr	February 2025

Indicator	Result	Note / source year
Capacity utilisation (approx.)	≈ 34%	0.76 ÷ 2.21 (author's calculation)

CPCB figures cover only 21 notified categories. UN-based estimates put India's e-waste at about 3.2 Mt, so use CPCB numbers for compliance and UN numbers for international comparison.

5. SWOT Analysis

Strengths

- Legally binding, rising recycling targets
- Digital registration and certificate tracking
- Includes refurbishers and solar PV waste
- Formalisation of the informal sector is an explicit goal

Weaknesses

- Weak enforcement and thin state pollution board capacity
- Registered recycling capacity is largely idle (≈ 34% used)
- Focus on recycling, with limited attention to repair
- Risk of paper certificates without real recycling

Opportunities

- Urban mining of gold, copper and rare earths
- Growth of domestic electronics manufacturing creating recovery demand
- Digital verification (e.g., GST e-invoice links used for battery EPR from July 2026)

Threats

- Litigation by producers over floor prices (April 2025)
- Informal buyers paying more in cash than formal recyclers
- Lithium-ion battery fires and illegal imports

6. Comparison with India

Table 3.3 compares the 2016 and 2022 Rules and gives a benchmark from the EU and Switzerland.

Table 3.3 Rules 2016 vs 2022 vs benchmark

Feature	India 2016 Rules	India 2022 Rules	EU / Switzerland
Approval	State-level authorisation	Central portal registration	National producer registers
EPR mechanism	Producers or PROs; collection targets	Recycling targets of 60% to 80%; tradable certificates	EU: collect 65% of EEE sold or 85% of WEEE generated; Swiss pooled ARF
Scope	21 product types	106 product types plus solar PV	Open scope across six categories (EU)
Refurbishers	Not in the EPR system	Certificates defer EPR	Reuse counted towards targets (EU)
Enforcement	Mainly state boards; weak	Compensation, audits, portal tracking; still evolving	National authorities; audited schemes

Key takeaway: The 2022 Rules close many design gaps of the 2016 Rules; the remaining gap is in enforcement and in moving money to collection.

7. Recommendations

1. Tighten verification with end-product tracking, weight-in/weight-out reconciliation, third-party audits and digital invoice links.
2. Bring informal collectors in as registered collection agents with a fair per-kg price.
3. Review the floor price of certificates regularly with transparent methodology, to reduce disputes.
4. Add repair, refurbishment and right-to-repair incentives so that products stay in use longer.
5. Strengthen state pollution control boards with staff, inspection targets and public dashboards.
6. Reconcile CPCB and UN estimates so that targets rest on credible baseline data.

Sources: [14] [15] [16] [17] [18] [19] [22] (see References)

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4. India's Extended Producer Responsibility (EPR): Success or Paper Compliance?

1. Executive Summary

EPR makes producers responsible for the end-of-life management of what they sell. India uses it for plastic packaging, e-waste, batteries and other streams, with a certificate market run by CPCB. The verdict on evidence so far is a **partial success**: EPR has created legal obligations, registration and a growing formal market, but audits found large-scale fake certificates and field evidence shows that money often rewards documents, not the physical collection of waste.

2. Background

EPR first appeared in Indian rules in 2011 (plastic and e-waste). The modern version relies on a CPCB portal where producers, importers and brand owners (PIBOs) register and buy certificates from registered recyclers. In 2022-23 about 18,000 plastic-packaging companies were registered with CPCB, with a combined target of roughly 3.3 Mt of recycling.

In 2023 an audit revealed that four firms in Karnataka, Maharashtra and Gujarat had issued nearly 600,000 fake plastic EPR certificates, and CPCB began auditing about 800 plastic recyclers. This incident is the main evidence for the 'paper compliance' concern.

3. Strategy / Initiative

- **Step 1: Obligation.** PIBOs register on the portal and receive annual targets by weight.
- **Step 2: Processing.** Registered recyclers collect and process material and upload records of end-product output.
- **Step 3: Certificates.** Verified output generates EPR certificates, which producers buy on a CPCB trading platform to offset their obligation.
- **Step 4: Oversight.** CPCB and state boards audit records and levy environmental compensation for shortfalls or fraud.
- **Step 5: Price support.** A floor price for e-waste certificates (reported at about ₹22/kg after a 2024 amendment) was introduced to keep formal recycling viable.

4. Impact & Results

Table 4.1 collects evidence for and against the success of EPR.

Table 4.1 EPR in India: evidence

Indicator	Result	What it suggests
E-waste share formally processed	26.3% (2020-21) → 32.9% (2021-22) → ≈ 43% (2023-24)	Formal share is rising
Recycler capacity vs. processed	2.21 Mt capacity vs ≈ 0.76 Mt processed	Plants idle for lack of feedstock
Plastic EPR registrations	≈ 18,000 companies; ≈ 3.3 Mt target (2022-23)	Wide coverage of producers
Fake plastic certificates found	≈ 600,000 from four firms; ≈ 800 recyclers audited	Verification failure
E-waste certificate prices	Reported ₹10–50/kg (2024-25); floor ≈ ₹22/kg	Price signal is weak and contested
Legal challenge	Major appliance makers challenged floor price in Delhi High Court (April 2025)	Producer resistance

Indicator	Result	What it suggests
Compliance cost example	Bajaj Electricals: ≈ ₹9.5 crore EPR (FY25), expected ≈ ₹18 crore next year	Costs rise as targets step up
Field evidence (Delhi, 2026)	Certificate markets pay for paperwork; cash-paying collectors outbid formal recyclers	Money does not reach collection

Figures come from CPCB-based summaries, company statements and press analyses; treat certificate prices as indicative.

5. SWOT Analysis

Strengths

- Legal backing and a national online portal
- Market flexibility: producers can choose the cheapest certificates
- Has pulled many producers and recyclers into registration

Weaknesses

- Certificates monetise documentation rather than collection
- Verification is weak; fake certificates were found
- Informal collectors pay more in cash than formal recyclers can
- Idle formal capacity

Opportunities

- Link EPR to GST e-invoices and digital tracking (already applied to battery EPR from July 2026)
- Use QR codes and product IDs for traceability
- Turn informal collectors into registered agents

Threats

- Loss of credibility if fraud continues
- Price collapse or floor-price litigation
- Producers treating EPR only as a cost, not a design driver

6. Comparison with India

Because this case is itself Indian, Table 4.2 benchmarks India's EPR against the Swiss and Japanese systems from Cases 1 and 2.

Table 4.2 EPR models compared

Feature	India (EPR certificates)	Switzerland (ARF)	Japan (appliance law)
Organiser	Producers or PROs via CPCB portal	Industry non-profits (SENS, Swico)	Producers and retailers under statute
Money flow	Producer buys certificate from recycler	Fee at sale goes into a pooled fund	Consumer pays fee at disposal plus producer costs
Verification	Portal records and audits (evolving)	Annual audits; export permits	Manifest (tracking-slip) system
Collectors	Informal buyers outbid formal ones	Retailers and municipal points, free return	Retailers and municipalities
Outcome	≈ 43% formally recycled (2023-24)	≈ 11–15 kg per person collected	Regulated appliance types widely recycled

> **Key takeaway:** India copied the idea of producer responsibility but not the controls: pooled funding, retailer take-back and audited flows.

7. Recommendations

1. Shift from 'paper' to 'physical' verification: mass-balance checks, geo-tagged manifests and GST-linked e-invoicing for all EPR streams.
2. Pay at the point of acquisition: allow part of the certificate value to fund per-kg collection payments so formal recyclers can bid for waste.
3. Register informal collectors as authorised agents with IDs, safe-handling training and fair prices.

4. Introduce graduated penalties, blacklisting of fraudulent recyclers and independent third-party audits.
5. Publish a public dashboard of certificates, targets met and audit outcomes.
6. Consider deposit-refund schemes for high-value items such as phones and laptops.

Sources: [18] [19] [20] [21] [23] [24] [25] [26] (see References)

CASE STUDY 5 OF 10

5. Bengaluru: India's E-Waste Capital?

1. Executive Summary

Bengaluru is India's IT hub and one of its top three e-waste-generating cities, but it is not the largest: rankings based on Assocham and CPCB data place Mumbai first and Delhi second. The city's real distinction is the mix of heavy corporate e-waste, a proactive state pollution board, home-grown recyclers such as E-Parisara, and a persistent informal economy in areas like Gowripalya and Nayandahalli. An industry official has put the formal share at only 10–15%.

2. Background

Bengaluru's IT and IT-enabled services sector generates large volumes of computers, servers and telecom equipment. Published estimates differ widely: an Assocham-based report cited about 20,000 t a year (growing about 20% annually); the state pollution board said about 1,000 t a month comes from IT companies alone; and a 2016 study cited in the press put Bengaluru at about 92,000 t. Experts also note that there is no mechanism to measure e-waste generation in the city accurately.

So the honest answer to the question in the case title is: Bengaluru is a top-three generator and a leader in *IT* e-waste and e-waste policy experiments, but 'capital' by volume belongs to Mumbai and Delhi.

3. Strategy / Initiative

- **Regulation and enforcement:** the Karnataka State Pollution Control Board (KSPCB) issued licences to about 150 dismantling, refurbishing and recycling units (only 86 operational in 2022) and repeatedly warned bulk consumers to comply with the rules.
- **Public collection bins:** in February 2022 KSPCB announced 50 e-waste bins across Bengaluru (at places such as Vidhana Soudha, the High Court and malls) and 100 more elsewhere in the state.
- **PROs and social enterprises:** organisations such as Saahas Zero Waste, BinBag and Hasirudala channel e-waste from firms and households to authorised recyclers.
- **Technology development:** MeitY funded a demonstration plant in Bengaluru (C-MET Hyderabad with E-Parisara, ₹1,126.8 lakh over five years from August 2014) for recovering metals from printed circuit boards.
- **Awareness drives:** KSPCB-run statewide e-waste awareness campaigns.

4. Impact & Results

Table 5.1 gathers the available (and inconsistent) numbers.

Table 5.1 Bengaluru: e-waste indicators

Indicator	Result	Note / source
City rank by e-waste generation	3rd (Mumbai 1st, Delhi 2nd)	Assocham/CPCB-based ranking
Annual generation (estimate 1)	≈ 20,000 t, rising ≈ 20%/yr	Assocham-based report
E-waste from IT firms	≈ 1,000 t per month	KSPCB
Annual generation (estimate 2)	≈ 92,000 t	2016 figure cited by press
Licensed units / operational	≈ 150 / 86	KSPCB, 2022
Public e-waste bins	50 in the city	KSPCB, 2022
Formal-sector share	≈ 10–15%	Industry official, ten years after the rules
Informal hubs	Gowripalya, Nayandahalli	Rajya Sabha report cited by Citizen Matters

The spread between 20,000 t and 92,000 t shows a data problem: Bengaluru lacks a reliable e-waste inventory.

5. SWOT Analysis

Strengths	Weaknesses
• Large, organised corporate generators that can be regulated • Active state board and a cluster of recyclers, PROs and start-ups • Local R&D on PCB metal recovery	• IT firms often auction e-waste to the highest bidder, usually an informal dismantler • Only 86 of 150 licensed units were operational • Low household awareness; low volumes in public bins • No reliable measurement of generation
Opportunities	**Threats**
• Corporate ESG commitments and buy-back or refurbishment for schools • Registered-aggregator model for informal hubs • Use of EPR money for collection	• Volume growing about 20% a year • Toxic burning and open dumping in informal hubs • Price competition from informal buyers

6. Comparison with India

Table 5.2 compares Bengaluru with the national picture and with other hotspots.

Table 5.2 Bengaluru vs India

Parameter	Bengaluru / Karnataka	India / other cities
City rank	3rd by generation	Mumbai 1st, Delhi 2nd
Formal share	≈ 10–15% (industry view)	≈ 33% (2021-22) and ≈ 43% (2023-24) of estimated national generation processed formally (not strictly comparable)
Informal hubs	Gowripalya, Nayandahalli	Seelampur and Mustafabad (Delhi), Moradabad (UP), Bhiwandi (Maharashtra)
Main generators	IT and corporate bulk consumers	Mix of households, businesses and institutions
Public infrastructure	50 KSPCB bins (2022)	Public e-waste bins are rare in most cities

Key takeaway: Bengaluru's challenge is less about volume than about controlling where corporate e-waste goes.

7. Recommendations

1. Enforce the rule that bulk consumers sell or auction e-waste only to registered recyclers, verified on the CPCB portal before each auction.
2. Expand public bins and doorstep pickup with per-kg buy-back payments so that formal channels match kabadiwala prices.
3. Register Gowripalya and Nayandahalli aggregators as authorised collectors, with safe-processing upgrades and worker training.
4. Build a city e-waste inventory and public dashboard.
5. Encourage IT firms to refurbish and donate usable equipment before recycling.
6. Continue support for local R&D on printed-circuit-board metal recovery.

Sources: [27] [28] [29] [30] [31] [36] (see References)

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6. Mumbai's E-Waste Ecosystem

1. Executive Summary

Mumbai is ranked first among Indian cities for e-waste generation, and Maharashtra is usually listed among the top generating states. The city's e-waste passes through a layered ecosystem: households and offices hand items to scrap dealers, aggregators trade them, a small formal recycling sector handles bulk and EPR volumes, and valuable components such as printed circuit boards travel to informal hubs elsewhere. In July 2026 the BMC house approved a proposal to frame a dedicated e-waste policy.

2. Background

A 2007 Toxics Link study estimated that Mumbai generated about 19,000 t of waste electrical and electronic equipment a year (Delhi about 11,000 t) and found that much of it left the city through informal trade, with circuit boards sent to Delhi NCR. A 2007 MPCB assessment put Greater Mumbai at 11,017 t out of 20,271 t for Maharashtra.

More recent official data reported by MPCB show Maharashtra's e-waste generation at 37,099 t in 2020-21 and 83,086 t in 2021-22. The sharp jump largely reflects changes in estimation methods, so it should be read cautiously. As a share of CPCB's national totals (1.35 Mt and 1.60 Mt), Maharashtra accounts for about 2.8% and 5.2%, which is lower than older studies suggested.

3. Strategy / Initiative

- **Generators:** households, offices, banks, IT firms, ports and industries. Most items go to kabadiwalas and local scrap dealers because they pay cash immediately.
- **Aggregators and informal dismantlers:** they strip out valuable parts; circuit boards and other fractions move through informal trade routes to hubs such as Delhi NCR and Bhiwandi.
- **Formal sector:** MPCB-authorised dismantlers, recyclers and PROs handle bulk consumer and EPR volumes. MPCB has treated e-waste as a priority since 2005 and initiated a Mumbai–Pune work group.
- **Municipal action:** the BMC house approved a proposal (July 2026, per media reports) for a dedicated e-waste management policy.
- **Related models:** SWaCH in Pune was authorised by MPCB in 2012 to collect and channel e-waste (see Case 8).

4. Impact & Results

Table 6.1 collects the main data points; older figures are shown with their year.

Table 6.1 Mumbai and Maharashtra: e-waste indicators

Indicator	Result	Note / source year
Rank among Indian cities	1st, then Delhi, Bengaluru, Chennai, Kolkata, Ahmedabad, Hyderabad, Pune, Surat, Nagpur	Assocham/CPCB-based ranking
Mumbai WEEE generation	≈ 19,000 t per year	Toxics Link, 2007
Greater Mumbai WEEE	11,017 t of 20,271 t (Maharashtra)	MPCB, 2007
Maharashtra generation	37,099 t (2020-21); 83,086 t (2021-22)	MPCB, via Lok Sabha reply
Share of India's total	≈ 2.8% (2020-21); ≈ 5.2% (2021-22)	Author's calculation from CPCB totals
Projection for Mumbai region	≈ 2.92 lakh t by 2020	Study projection, not observed data

Indicator	Result	Note / source year
Policy	BMC to frame e-waste policy	House approval, July 2026

Different studies use different methods, so numbers from different years should not be compared directly.

5. SWOT Analysis

Strengths	Weaknesses
• Large market and dense collection network of scrap dealers • Presence of authorised recyclers and PROs • MPCB and BMC attention to the issue	• Most e-waste is handled informally • Lack of dedicated municipal collection or public bins • Weak data: estimates vary widely
Opportunities	**Threats**
• City policy can integrate scrap dealers as authorised collectors • Retail and corporate take-back partnerships • Link with waste-picker cooperatives (Pune model)	• Illegal imports and clandestine trade • Toxic burning and dump fires • Exporting the problem to informal hubs in other regions

6. Comparison with India

Table 6.2 compares Mumbai with India and with Pune/Kerala models.

Table 6.2 Mumbai vs India

Parameter	Mumbai	India / benchmark
Rank	1st among cities	Top cities: Delhi, Bengaluru, Chennai, Kolkata
Collection	Largely through scrap dealers	Kerala: Haritha Karma Sena; Pune: SWaCH bins
Formal infrastructure	Authorised recyclers and PROs; limited public bins	National: 322 registered recyclers (2025)
Data quality	Old studies plus MPCB totals	CPCB national inventory (21 categories)
City policy	In preparation (2026)	Some states run statewide drives (Kerala 2025)

> **Key takeaway:** Mumbai has scale and formal capacity; what it lacks is a municipal collection system that competes with cash buyers.

7. Recommendations

1. Frame the BMC e-waste policy around collection: public bins, doorstep pickup, ward-level collection days and buy-back rates.
2. Register scrap dealers and kabadiwalas as authorised collection agents with ID cards and training.
3. Track flows out of the city, including circuit boards leaving for other states, through manifests.
4. Partner with malls, housing societies and offices for take-back points.
5. Commission a fresh, methodologically consistent city inventory.

Sources: [32] [33] [34] [35] [36] [37] [38] (see References)

CASE STUDY 7 OF 10

7. Kerala's Clean Kerala Company

1. Executive Summary

Clean Kerala Company Limited (CKCL) is a Government of Kerala company that acts as the state's collector-aggregator for non-biodegradable waste and e-waste. It works with Haritha Karma Sena (HKS), the Kudumbashree women's green workforce that collects door to door and charges a user fee. Collected material goes to Material Collection Facilities and then to CKCL, which sorts it and passes it to recyclers. HKS delivered 50,190 t of non-biodegradable waste to CKCL in 2024-25 and 86,296 t in 2025-26, and a statewide household e-waste drive with per-kg payments started in July 2025.

2. Background

Kerala is densely populated, highly literate and a high consumer of electronics, which makes open dumping hard to sustain. The state built a decentralised system around local bodies, the Suchitwa Mission, the Haritha Keralam Mission and Kudumbashree.

CKCL began taking institutional e-waste in 2014 at ₹5 per kg, sending it to a recycler in Palakkad, and signed pacts with municipalities such as Kochi. An early phase collected about 108 t in six months. Haritha Karma Sena, launched as a Kudumbashree initiative from 2017, now works in every local body.

3. Strategy / Initiative

- **Door-to-door collection by HKS:** trained green technicians (mostly women) collect non-biodegradable waste and charge a user fee collected through local bodies.
- **Material Collection Facilities (MCFs):** waste is sorted at ward or panchayat level.
- **CKCL as aggregator:** the company collects sorted material from MCFs, pays HKS for it and hands it to recyclers.
- **Statewide e-waste drive (July 2025):** HKS collects 44 types of non-hazardous e-waste from households and institutions and pays a price per kg from a consortium fund or local body funds; the amount is later reimbursed by CKCL.
- **Livelihood diversification:** HKS units also run cloth-bag and paper-bag making units from waste.

4. Impact & Results

Table 7.1 shows the scale of the system.

Table 7.1 Kerala: Haritha Karma Sena and CKCL results

Indicator	Result	Note / source year
Non-biodegradable waste handed to CKCL	50,190 t	2024-25
Non-biodegradable waste handed to CKCL	86,296 t (+ ≈ 72%)	2025-26, highest so far
HKS units and women workers	4,438 units; 35,214 women (2024-25); 37,481 women (2025-26)	Kudumbashree / LSGD
User fee collected by HKS	≈ ₹341 crore	2024-25
Payment for sorted waste	≈ ₹7.8 crore	2024-25
Statewide household e-waste drive	33,945 kg in first month; ₹2.63 lakh paid	July–August 2025
Bag-making units run by HKS	223 cloth-bag units; 540 paper-bag units	2024-25

Sources give different totals for 2024-25 (50,190 t by Kudumbashree; over 66,000 t by the minister) because their scope differs.

5. SWOT Analysis

Strengths	Weaknesses
• Decentralised, women-led workforce with a stable user-fee model • State-owned aggregator linking local bodies to recyclers • Household e-waste now priced per kg	• E-waste drive is new; volumes are small compared with likely generation • Depends on recyclers in other states • Fee collection and household compliance vary by local body
Opportunities	**Threats**
• Link CKCL flows with producer EPR money • Reverse logistics and refurbishing hubs • Replicating the model in other states (Tamil Nadu is adopting similar practices)	• Scrap dealers paying more in cash • Fiscal dependence on local bodies and government funding • Inconsistent data and metrics

6. Comparison with India

Table 7.2 compares Kerala with a typical Indian city and with Pune.

Table 7.2 Kerala vs India

Parameter	Kerala (HKS + CKCL)	Typical Indian city / Pune
Collection	Statewide door-to-door HKS in every local body	Contractor-based or partial; Pune's SWaCH covers about 70% of the city
Workforce	35,000+ women, organised through Kudumbashree	Waste pickers often informal; Pune: cooperative of ≈ 3,500
Financing	User fee plus payment for sorted waste	Municipal contracts; Pune: user fees ₹45–130/month
E-waste	Statewide per-kg drive from July 2025	Mostly via scrap dealers (Mumbai, Bengaluru)
Aggregator	State company (CKCL)	Typically none; Pune uses municipal feeder points

> **Key takeaway:** Kerala shows that a state-level institutional link between local workers and recyclers can scale collection quickly.

7. Recommendations

1. Sign EPR agreements so that producer money reimburses CKCL and HKS for e-waste collected.
2. Raise per-kg e-waste prices to match scrap-dealer offers and publicise them.
3. Build in-state dismantling and refurbishing capacity to cut transport and dependence on other states.
4. Standardise data and publish a single dashboard of tonnage, payments and destinations.
5. Strengthen user-fee collection, linking it to property tax as the state is planning.

Sources: [39] [40] [41] [42] [43] [44] (see References)

CASE STUDY 8 OF 10

8. Pune's SWaCH Model

1. Executive Summary

SWaCH is a wholly owned cooperative of waste pickers that provides door-to-door waste collection in Pune under an agreement with the Pune Municipal Corporation (PMC). Households pay a small monthly user fee, waste pickers earn from fees and from selling recyclables, and the city saves on transport and processing. SWaCH reports about 3,500 members, about 220 t a day sent for recycling and savings to the city of over ₹100 crore a year, though workers' earnings and conditions remain a concern.

2. Background

Before 2005, Pune residents dumped waste in public containers, informal pickers scavenged recyclables, and almost no segregation took place. Waste grew from about 300 t a day in 1991 to about 1,700 t a day in 2016. The waste-pickers' union KKPKP organised a 2005 pilot with SNDT University in which about 1,500 pickers provided door-to-door collection to about 150,000 households.

SWaCH Pune Seva Sahakari Sanstha was formed in 2007, and PMC signed its first five-year agreement in 2008. In 2012 the Maharashtra Pollution Control Board authorised SWaCH to collect and channel e-waste.

3. Strategy / Initiative

- **Service contract:** PMC provides ID cards, infrastructure and feeder points; SWaCH members collect segregated waste from doorsteps (each picker serves roughly 150–200 households).
- **User fee model:** households pay directly. In 2019 the fees were ₹65 (residential), ₹130 (commercial) and ₹45 (slum) per month, revised to ₹70, ₹140 and ₹50 in 2020, with a ₹10 slum subsidy per property.
- **Extra income:** members may sell recyclables from dry waste to scrap dealers and sort unsegregated waste.
- **Helpline and accountability:** households can complain about missed pick-ups.
- **Zero-waste integration:** composting in housing societies, sanitary-waste 'red dot' campaign and EPR-linked collection of low-value plastics.

4. Impact & Results

Table 8.1 shows the reported impacts.

Table 8.1 Pune SWaCH: impact data

Indicator	Result	Note / source
Waste pickers in cooperative	≈ 1,500 (2005 pilot) → ≈ 3,500 (2020)	SWaCH / PMC presentation
City coverage	≈ 70% of the city (2020); ≈ 400,000 households earlier	PMC, Down To Earth
Waste sent for recycling	≈ 220 t/day (2020); 227.1 t/day (82,891.5 t/yr)	PMC/SWaCH; Press Institute
Segregation reported	98%	Self-reported in 2020 presentation
Savings to city	≈ ₹113 crore a year vs cost of ≈ ₹8 crore; contractors would cost ₹180–290 crore	PMC/SWaCH presentation (claimed)
Plastic diverted	≈ 30,000 t a year; 52% of Pune's plastic waste	UNESCAP

Indicator	Result	Note / source
Emissions avoided	≈ 50,000 t CO2e per year (SWaCH claims up to 100,000 t)	UNESCAP; SWaCH

Many figures are self-reported by SWaCH or PMC; independent audits are limited.

5. SWOT Analysis

Strengths

- Dual benefit: better service and dignified work for waste pickers
- Low cost for the municipality compared with contractors
- Doorstep segregation and trust built with households
- Cooperative ownership and union support

Weaknesses

- Fee non-payment and low fee levels
- Incomes remain modest; 'green' but not always 'decent' work
- Dependence on PMC contract renewal

Opportunities

- EPR-linked payments for dry-waste collection
- Decentralised composting and e-waste handling
- Replication in other cities and states

Threats

- Push for large integrated contractors and mechanised or waste-to-energy plants
- Political and contractual uncertainty
- Rising labour and transport costs

6. Comparison with India

Table 8.2 compares SWaCH with a typical Indian city system.

Table 8.2 SWaCH vs typical Indian city

Parameter	Pune SWaCH	Typical Indian city
Collection agent	Waste-picker cooperative	Private contractor or municipal staff
Segregation	98% reported (2020)	Mixed waste common
Informal workers	Included, with ID cards and fees	Often excluded and unpaid
Cost to city	≈ ₹8 crore a year (claimed)	Contractor estimates ₹180–290 crore a year for equivalent service (claimed)
Recyclables	Recovered by workers at the source	Lost to landfill or recovered by scavenging at dumps

Key takeaway: SWaCH shows that formalising waste pickers is both fairer and cheaper than displacing them.

7. Recommendations

1. Make waste-picker cooperatives the default door-to-door partner in every city, as recommended for SWM Rules implementation.
2. Fix a minimum user fee and tie its collection to property tax to secure worker incomes.
3. Channel EPR funds to cooperatives for dry-waste and e-waste collection.
4. Provide health insurance, protective equipment and pensions.
5. Commission independent audits to validate savings and diversion claims.

Sources: [45] [46] [47] [48] [49] [50] (see References)

CASE STUDY 9 OF 10

9. Dharavi Recycling Ecosystem

1. Executive Summary

Dharavi in Mumbai is one of Asia's largest informal recycling clusters. Thousands of small units sort, wash, shred and trade plastics, metals, paper, cardboard, glass and e-waste, supported by up to hundreds of thousands of waste pickers across the city. Estimates say it handles 50–80% of Mumbai's waste and about 60% of its plastic, though there are no official numbers. It is a low-cost, high-employment recycling engine, but with unsafe conditions and now a redevelopment threat.

2. Background

Dharavi dates to 1882, when polluting industries were moved to the site. Today it occupies roughly 600 acres in central Mumbai and houses more than a million people. Recycling gradually became a profitable business as Mumbai's waste grew; more than 5,000 single-room units sort and process waste, concentrated around the '13th Compound'.

Reported annual turnover ranges from about US$650 million to US$1 billion, with some observers claiming much more. Workers such as sorters may earn about ₹500 a day and waste pickers ₹300–350 a day.

3. Strategy / Initiative

- Dharavi's recycling system is an emergent, market-driven ecosystem rather than a planned initiative. Its value chain has five links:
- **Collection:** waste pickers and small vendors collect from across Mumbai, with up to about 250,000 livelihoods depending on the chain.
- **Sorting:** units grade plastics into several types (a worker interviewed in 2026 described seven grades), metals, paper and glass.
- **Processing:** washing, shredding, melting and pelletising of plastics; metal melting; e-waste dismantling.
- **Trading:** materials sold to manufacturers in Mumbai and elsewhere.
- **Support activities:** leather, textiles and other small manufacturing in the same cluster.
- **Professionalisation efforts:** NGOs such as Earth5R train workers in safer processing, while the Dharavi Redevelopment Project (Adani Group-linked) proposes to rehouse residents and has raised concerns for the recycling units.

4. Impact & Results

Table 9.1 summarises the estimates. All are approximate.

Table 9.1 Dharavi: recycling estimates

Indicator	Estimate	Note / source
Area and population	≈ 600 acres (≈ 175–240 ha); over 1 million residents	Sources vary
Recycling units	5,000+ single-room units	Earth5R / Acorn
Share of Mumbai's waste recycled	50–80% (estimates vary); ≈ 60% of plastic	No official numbers
Livelihoods	Up to ≈ 250,000 (incl. city-wide pickers); ≈ 40,000 in units	Pulitzer/Maktoob 2026; Tomorrow.city
Annual turnover	US$650 million to US$1 billion (some claim more)	Dialogue Earth; Revolve
Typical daily earnings	₹300–350 (picker); ₹500 (sorter)	2026 reporting

Exam tip: always say 'estimated'; no census of Dharavi's recycling economy exists.

5. SWOT Analysis

Strengths	Weaknesses
• Very low-cost, skilled and dense value chain • High employment and livelihood support • Diverts large amounts of plastic from landfills	• Unsafe working conditions, burning and health risks • No legal recognition or formal data • Low and insecure wages
Opportunities	**Threats**
• In-situ upgrade with safe workshops and worker cooperatives • Integration into EPR chains as authorised recyclers • Skilling and finance for units	• Displacement by redevelopment and high land values • Formal-sector capture of the waste stream • Fires, plastic bans and regulatory crackdowns

6. Comparison with India

Table 9.2 compares Dharavi with Pune's semi-formal SWaCH model and formal systems abroad.

Table 9.2 Three system types

Parameter	Dharavi (informal)	Pune SWaCH (semi-formal)	Japan / Switzerland (formal)
Legal status	Unregistered or partly registered	Cooperative with municipal contract	Registered and audited
Worker protection	Minimal	ID cards, some benefits	Full labour protections
Data	Estimates only	Self-reported by SWaCH/PMC	Detailed annual reports
Cost	Very low	Low	High
Environmental safety	Poor	Moderate	High

Key takeaway: Dharavi shows the recycling capacity of the informal sector; Pune shows how to formalise it without destroying it.

7. Recommendations

1. Recognise Dharavi's recycling units through registration, waste-picker ID cards and access to EPR contracts.
2. Protect the recycling economy in any redevelopment by reserving in-situ, purpose-built industrial space for existing units.
3. Improve safety through shared effluent treatment, fire safety and protective equipment.
4. Carry out a proper census of units, workers and material flows to replace estimates.
5. Offer low-cost finance and skills training so units can meet formal quality standards.

Sources: [51] [52] [53] [54] [55] [56] (see References)

CASE STUDY 10 OF 10

10. Dell's Circular Economy Strategy

1. Executive Summary

Dell Technologies has set three 2030 circular-economy goals: reuse or recycle one metric ton of product for every metric ton sold, make 100% of packaging from recycled, renewable or reused material, and make more than half of product content from recycled, renewable or reduced-carbon materials. Its strategy combines design for repair and disassembly, recycled and bio-based materials, closed-loop plastics and aluminium, take-back and asset recovery, and partnerships such as Goodwill and the Circular Electronics Partnership. Reported packaging progress was 90.2% in FY2022; the figures are company-reported.

2. Background

Dell describes more than 25 years of responsible recycling services for customers. In November 2019 it published its 'Progress Made Real' 2030 plan, and in 2021 it joined the Circular Electronics Partnership. The global context is stark: the world produced about 62 million tonnes of e-waste in 2022 and only around 22% was documented as properly collected and recycled.

3. Strategy / Initiative

- **Design:** modular, repairable products and prototypes (such as Concept Luna) that make parts easy to access and reuse; spare parts and repair tutorials to extend life.
- **Materials:** recycled aluminium, recycled steel, copper and cobalt; post-consumer recycled plastics; ocean-bound plastic; reclaimed carbon fibre from aerospace; bio-based plastics; and closed-loop use of plastics recovered from returned e-waste.
- **Packaging:** bamboo, recycled paper pulp and sugarcane fibre replacing foam and plastic; commercial laptop packaging is now made from recycled or renewable materials.
- **Take-back and asset recovery:** returned devices are assessed; those that can be repaired are refurbished and resold, and the rest are dismantled for parts and materials.
- **Partnerships and reporting:** Goodwill (through the Dell Reconnect programme), the Circular Electronics Partnership, and annual ESG progress reports.

4. Impact & Results

Table 10.1 lists the goals and reported progress. All data are company-reported.

Table 10.1 Dell: circular economy goals and progress

Indicator	Result	Note / source year
Goal 1: take-back	1 t reused/recycled per 1 t sold by 2030	Goal
Goal 2: packaging	100% recycled, renewable or reused by 2030	Goal; 85% (2021) → 90.2% (FY2022)
Goal 3: product content	More than 50% recycled, renewable or reduced-carbon by 2030	Goal
Sustainable materials used	≈ 396.5 million lb (≈ 179.8 million kg) in products and packaging	FY2022
Products taken back for reuse/recycling	+26% year on year (share of products sold)	FY2022
Material recovered via Goodwill	≈ 620 million lb (cumulative)	Over 2,000 Goodwill locations, FY2023
Ocean-bound plastic	443,000+ lb kept out of oceans; 227,595 kg used in packaging	Dell website; FY2022

Progress data are self-reported, not independently audited; check Dell's latest ESG report before quoting numbers.

5. SWOT Analysis

Strengths	Weaknesses
• Scale and supply-chain influence • Closed-loop material innovation • Long-standing take-back programmes and partnerships • Public goals and annual reporting	• Goal 3 counts 'reduced-carbon' materials, a broad definition • 1:1 take-back is measured by weight, not by the same devices • Progress relies on the supply of recycled feedstock
Opportunities	**Threats**
• Growing demand for repair and refurbished devices • Rising regulation (EPR, ecodesign) rewards early movers • Growth markets such as India for take-back services	• Greenwashing criticism if data are not audited • Cost and availability of recycled materials • Short device life and consumer preference for new models

6. Comparison with India

Table 10.2 compares Dell's voluntary goal with India's EPR requirements and typical producer practice.

Table 10.2 Dell vs India's EPR requirements

Parameter	Dell (voluntary goal)	India (E-Waste Rules 2022)
Take-back ambition	1 t reused/recycled per 1 t sold by 2030 (100%)	EPR target 60% (2023-24) rising to 80% (2027-28)
Method	Own programmes and partners	Buy EPR certificates from registered recyclers
Design	Design for repair, recycled content, packaging	Hazardous-substance limits; no general recycled-content mandate for EEE
Transparency	Annual ESG report (self-reported)	CPCB portal; limited public data
Producer stance	Marketing and regulatory advantage	Some producers contest floor prices (2025 litigation)

Key takeaway: A voluntary corporate goal of 100% by weight already exceeds India's legal target; the gap is in design and transparency.

7. Recommendations

1. Companies in India should publish independently audited circularity data and their EPR fulfilment.
2. Offer doorstep take-back with trade-in or buy-back incentives to beat the kabadiwala price.
3. Extend repair and refurbishing services and support the right to repair.
4. Policymakers can add recycled-content and design-for-repair requirements to EEE rules, learning from Dell's practices.
5. Define terms precisely (recycled, renewable, reduced-carbon) to prevent greenwashing.

Sources: [57] [58] [59] [60] [61] [62] [63] (see References)

11. Conclusion

The ten cases show that there is no single route to high recycling. Japan and Switzerland reach high performance through formal, rule-bound systems in which every party has a defined role and the money is collected in advance. India's own examples show a different path: Pune's SWaCH and Kerala's Haritha Karma Sena succeed by paying and organising the people who already collect waste, while Dharavi shows how much the informal sector can do without any support. Bengaluru and Mumbai show what happens when corporate and household e-waste is left to the cash market. India's E-Waste Rules 2022 and the wider EPR system supply the legal framework, and Dell shows what a producer can do when it treats circularity as a design and business goal.

Table 11.1 Cross-case comparison of models

Model	Cases	Main strength	Main weakness
Formal, regulated	1 Japan; 2 Switzerland	High collection and recycling; clear roles; pre-funded	High cost; heavy reliance on citizen compliance; Japan relies on incineration
Policy framework	3 Rules 2022; 4 EPR	Binding targets; national portal; certificate market	Weak verification and enforcement; fake certificates; idle capacity
Urban hotspots	5 Bengaluru; 6 Mumbai	Large volumes and established recyclers	Informal handling; poor data; no municipal collection
Inclusive, decentralised	7 Kerala; 8 Pune	Fair work, low cost, high segregation	Fee collection; modest incomes; reliance on public contracts
Informal market	9 Dharavi	Low-cost, high-skill, high-employment	Unsafe work; no recognition; redevelopment risk
Corporate circularity	10 Dell	Design, materials and take-back at scale	Self-reported data; broad definitions

Five lessons for India

1. **Move the money to the collection point.** Certificates and fees must reach whoever gets the item from the household or company.
2. **Include the informal sector.** Pune, Kerala and Dharavi show that workers are the system's greatest asset.
3. **Verify physical flows.** Weight-in/weight-out records, audits and digital invoice links matter more than paper targets.
4. **Make return easy.** Free retailer take-back, public bins and doorstep pickup, as in Switzerland.
5. **Design and data.** Design for repair and recycling, and publish credible, independently checked numbers.

Data note: figures come from government replies, industry reports and press coverage retrieved in September 2026. Estimates differ by source and year; where they conflict, both are shown. Company figures (Dell) and cooperative figures (SWaCH) are self-reported.

12. References

Numbers match the citations at the end of each case study.

[1] Wikipedia. Recycling in Japan. https://en.wikipedia.org/wiki/Recycling_in_Japan

[2] IGES-CCET (M. Hayashi). PET bottle recycling in Japan (webinar slides). https://www.iges.or.jp/sites/default/files/inline-files/00_IGES_Miho_Hayashi_je-min.pdf

[3] Statista / Council for PET Bottle Recycling. Recycling rate of PET bottles in Japan. https://www.statista.com/statistics/695540/japan-pet-bottles-recycling-rate

[4] Japan for Sustainability (2005). Japan achieves high PET bottle collection rate at 61%. https://www.japanfs.org/en/news/archives/news_id025816.html

[5] PlasticsToday. Japan targets 100% PET recycling ratio by 2030. https://www.plasticstoday.com/advanced-recycling/japan-targets-100-pet-recycling-ratio-by-2030

[6] Dezeen (25 Jul 2019). Olympic committee unveils 2020 medals made from recycled smartphones. https://www.dezeen.com/?p=1387533

[7] SAYS (2019). The Tokyo 2020 Olympics medals are recycled from old mobile phones and other electronics. https://says.com/my/news/japan-is-using-recycled-materials-from-handphones-to-make-tokyo-2020-olympic-medals

[8] TriplePundit (2016). Tokyo 2020 medals may be made from e-waste. https://triplepundit.com/story/2016/2020-tokyo-summer-olympics-medals-may-be-made-e-waste/23216

[9] SENS eRecycling. Facts and figures 2025. https://www.erecycling.ch/en/ueber-sens/medien-und-publikationen/geschaeftsbericht/2025/facts-figures.html

[10] SENS eRecycling. Facts and figures 2023. https://erecycling.ch/en/wissenswertes/geschaeftsbericht/overview-2023/facts-figures.html

[11] SENS eRecycling. Facts and figures 2024 (mirror). https://erecycling.mironet.ch/en/wissenswertes/geschaeftsbericht/overview-2024/facts-figures.html

[12] SENS Foundation. SENS factsheet (WEEE Forum). https://weee-forum.org/wp-content/uploads/2019/03/SENS-factsheet_2023.pdf

[13] EUWID Recycling (2015). WEEE volumes collected in Switzerland increased by 4 per cent in 2014. https://www.euwid-recycling.com/news/business/weee-volumes-collected-in-switzerland-increased-by-4-per-cent-in-the-year-2014-050815/

[14] International Energy Agency. E-Waste Management Rules, 2022 (policy page). https://www.iea.org/policies/25027-e-waste-management-rules-2022

[15] Taxscan. E-Waste (Management) Rules 2022 to be effective from April 1, 2023 (Economic Survey 2023). https://www.taxscan.in/?p=250459

[16] TaxGuru. E-Waste Management Rules 2022 (Schedule III targets). https://taxguru.in/corporate-law/e-waste-management-rules-2022-1st-april-2023.html

[17] Toxics Link. Status of e-waste in the country (Parliament Q&A, CPCB data). https://toxicslink.org/?p=7576

[18] Toxics Link. Growing e-waste in the country (Parliament Q&A, Feb 2025). https://toxicslink.org/parliament-q-a/growing-e-waste-in-the-country/

[19] Tele.net.in. E-waste in India, 2023-24 recycling data. https://tele.net.in/?p=180341

[20] Drishti IAS (4 Oct 2025). Sustainable e-waste management strategies. https://www.drishtiias.com/daily-updates/daily-news-analysis/sustainable-e-waste-management-strategies/print_manually

[21] Drishti IAS (5 Aug 2025). Transforming e-waste management in India. https://www.drishtiias.com/daily-updates/daily-news-editorials/transforming-e-waste-management-in-india/print_manually

[22] IMPRI. E-Waste (Management) Rules (2022): assessing India's EPR framework. https://www.impriindia.com/?p=75001

[23] The Chatter (Zerodha). How India's e-waste rules found their teeth (EPR certificate prices and floor price). https://thechatter.zerodha.com/p/how-indias-e-waste-rules-found-their

[24] The India Forum (24 Sep 2026). Paper compliance, physical reality in e-waste management. https://www.theindiaforum.in/public-policy/paper-compliance-physical-reality-e-waste-management

[25] CivilsDaily. CPCB audit of plastic-waste recyclers (fake EPR certificates). https://civilsdaily.com/?p=769390

[26] Lawrbit. GST e-invoicing and the Battery Waste EPR portal. https://www.lawrbit.com/ehs/gst-e-invoicing-battery-waste-epr-portal/

[27] Citizen Matters. Between flames, fumes and bare hands: the untold story of Bengaluru's e-waste. https://citizenmatters.in/articles/the-untold-story-of-bengalurus-e-waste

[28] Deccan Herald. As e-waste piles up, Bengaluru IT firms slammed for 'time bomb'. https://www.deccanherald.com/amp/story/india%2Fkarnataka%2Fbengaluru%2Fas-e-waste-piles-up-bengaluru-it-firms-slammed-for-time-bomb-1054607.html

[29] Udayavani (11 Feb 2022). State pollution body to put 50 e-waste collection bins in Bengaluru. https://udayavani.com/?p=1213600

[30] Lok Sabha unstarred question AU3395 (MeitY demonstration plant, Bengaluru). https://eparlib.nic.in/bitstream/123456789/953436/1/AU3395.pdf

[31] The News Minute. Bengaluru to get India's first government-run e-waste recycling plant. https://www.thenewsminute.com/article/bengaluru-get-india-s-first-government-run-e-waste-recycling-plant-84097

[32] Down To Earth (2007), via India Environment Portal. Mumbai produces highest e-waste. https://indiaenvironmentportal.org.in/news-clippings/mumbai-produces-highest-e-waste

[33] Toxics Link. Mumbai: Choking on e-waste. https://toxicslink.org/wp-content/uploads/2022/05/Mumbai-Choking-E-waste_report_mail.pdf

[34] India Environment Portal. After plastic, govt aims a crackdown on e-waste in Maharashtra (MPCB 2007 assessment). https://indiaenvironmentportal.org.in/news-clippings/after-plastic-govt-aims-a-crackdown-on-e-waste-in-maharashtra

[35] Lok Sabha unstarred question AU2348 (CPCB and MPCB e-waste data). https://eparlib.nic.in/bitstream/123456789/2973285/1/AU2348.pdf

[36] Current World Environment 9(3). E-waste paper with city and state rankings. https://mail.cwejournal.org/pdf/vol9no3/vol9_no3_814-819.pdf

[37] Navbharat Live (Jul 2026). BMC to bring e-waste management policy. https://navbharatlive.com/maharashtra/mumbai/mumbai-e-waste-bmc-to-bring-management-policy-tejinder-singh-proposal-approved-1876822.html

[38] NSWAI. E-waste management in Mumbai Metropolitan Region: constraints and opportunities. https://www.nswai.org/docs/E-WASTE%20MANAGEMENT%20IN%20MUMBAI%20METROPOLITAN%20REGION%20-%20CONSTRAINTS%20AND%20OPPORTUNITIES.pdf

[39] The News Minute (30 Oct 2014). E-waste collection to begin in Kerala soon. https://www.thenewsminute.com/kerala/e-waste-collection-begin-kerala-soon-17209

[40] India Environment Portal. Kochi Corporation inks pact with Clean Kerala to dispose e-waste. https://indiaenvironmentportal.org.in/news-clippings/kochi-corporation-inks-pact-with-clean-kerala-to-dispose-e-waste

[41] LSG Kerala / Kudumbashree (26 Mar 2025). Kudumbashree Haritha Karma Sena: Kerala model of cleanliness. https://lsgkerala.gov.in/en/kudumbashree/news/kudumbashree-haritha-karma-sena-kerala-model-cleanliness

[42] The South First (15 Jul 2025). Kerala launches statewide e-waste collection drive. https://thesouthfirst.com/kerala/kerala-launches-statewide-e-waste-collection-drive/

[43] Kerala Kaumudi (Aug 2025). Haritha Karmasena's e-waste collection deserves praise. https://keralakaumudi.com/en/news/news-amp.php?id=1592515

[44] UNI (Aug 2026). Haritha Karma Sena collects 86,295.723 MT of non-biodegradable waste in 2025-26. https://test.uniindia.com/news/south/life-kerala-waste/3936373.html

[45] Centre for Public Impact. Waste management cooperative, Pune, India. https://www.centreforpublicimpact.org/case-study/waste-management-cooperative-pune-india

[46] SWaCH. SWaCH-PMC partnership. https://swachcoop.com/about/swach-pmc-partnership/

[47] Down To Earth. What a bright idea (SWaCH). https://www.downtoearth.org.in/waste/what-a-bright-idea--43782

[48] UNESCAP. Case study on the SWaCH model (repository). https://repository.unescap.org/bitstreams/f57dc371-1b0c-4124-bb79-91db407d52d6/download

[49] PMC / SWaCH presentation (CSE archive). https://cdn.cseindia.org/attachments/0.15083800_1626946321_pune-municipal.pdf

[50] Press Institute of India. 'Green' work, but not decent work (SWaCH). https://pressinstitute.in/?p=29388

[51] Dialogue Earth. Dharavi's recycling industry. https://dialogue.earth/en/?p=28790

[52] Tomorrow.city. Turning trash into treasure: inside Dharavi, Asia's largest slum. https://www.tomorrow.city/?p=33978

[53] Maktoob Media / Pulitzer Center (May 2026). Dharavi's recycling economy and what comes after. https://maktoobmedia.com/share/115538

[54] Feminism in India (18 Oct 2024). From slum to skyline: the implications of Adani's Dharavi redevelopment plan. https://feminisminindia.com/2024/10/18/from-slum-to-skyline-the-implications-of-adanis-dharavi-redevelopment-plan

[55] Revolve Media (18 Mar 2024). Circular // Views (Dharavi). https://revolve.media/?p=93933

[56] Acorn International. Dharavi recycling supply chain. https://acorninternational.org/category/acorn-india/page/5/

[57] Dell Technologies. Our ESG goals. https://www.dell.com/en-in/dt/corporate/social-impact/reporting/goals.htm

[58] Dell Technologies. Accelerating the circular economy. https://www.dell.com/en-au/lp/dt/circular-economy

[59] Dell Technologies. Sustainable devices. https://www.dell.com/en-us/lp/dt/sustainable-devices

[60] Dell Technologies blog. ESG at Dell Technologies: our purpose in action. https://dell.com/en-us/blog/esg-at-dell-technologies-our-purpose-in-action

[61] Waste360. What's inside Dell's sustainability playbook? https://www.waste360.com/sustainability/-what-s-inside-dell-s-sustainability-playbook-

[62] Business Today (21 May 2021). Committed to take back as much as we can: Dell bets big on recycling. https://businesstoday.in/technology/news/committed-to-take-back-as-much-as-we-can-dell-bets-big-on-recycling/story/439689.html

[63] Futurum Group. Dell Technologies making real progress on ESG and people-focused initiatives. https://futurumgroup.com/?p=19138

[64] TE_Case_studies.docx (course brief listing the ten case topics).

13. One-Page Exam Revision Sheet

Key numbers and one-line lessons for all ten cases

#	Case	Must-remember facts and numbers	Exam takeaway
1	**Japan recycling**	PET recycled ≈ 87% (FY22); 99.6% of municipalities collect PET; laws: packaging 1995, appliances 2001; overall recycling ≈ 20%, incineration > 70%; Tokyo 2020: 78,985 t e-waste → ≈ 5,000 medals (32 kg gold).	Clear roles + strict sorting + eco-design; but heavy incineration.
2	**Switzerland e-waste**	ORDEA 1998; ARF paid at sale; SENS + Swico ≈ 15 kg/person; SENS 2025: 104,751 t (11.48 kg); 634 points, 17 recyclers, 1,557 ARF payers.	Free return + pre-funded, audited, industry-run system.
3	**E-Waste Rules 2022**	Notified 2 Nov 2022, in force 1 Apr 2023; 106 items; targets 60/60/70/70/80/80%; EPR certificates on CPCB portal; 322 recyclers (2.21 Mt); 2023-24: ≈ 0.76 of 1.75 Mt recycled (≈ 43%).	Strong design; weak enforcement and verification.
4	**EPR: success or paper?**	≈ 600,000 fake plastic certificates (4 firms); ≈ 800 recyclers audited; e-waste floor price ≈ ₹22/kg (reported); OEM challenge Apr 2025; capacity utilisation ≈ 34%.	Partial success; verify physical flows; pay at collection.
5	**Bengaluru**	Top-3 city (Mumbai 1st, Delhi 2nd); ≈ 1,000 t/month from IT firms; formal share ≈ 10–15%; hubs Gowripalya, Nayandahalli; 150 licences / 86 operational; 50 bins (2022).	Corporate e-waste goes to highest bidder; enforce registered-recycler auctions.
6	**Mumbai**	Ranked 1st city; Toxics Link 2007: ≈ 19,000 t; Maharashtra 37,099 t (2020-21), 83,086 t (2021-22); PCBs to Delhi NCR; BMC e-waste policy (Jul 2026).	Scale is high; collection system is weak.
7	**Kerala CKCL**	CKCL + Haritha Karma Sena; 50,190 t (2024-25) → 86,296 t (2025-26); 4,438 units; ≈ 37,000 women; ₹341 crore fees; e-waste drive Jul 2025: 44 items, 33,945 kg in month 1.	Decentralised women workforce + state aggregator scales fast.
8	**Pune SWaCH**	KKPKP pilot 2005; coop 2007; PMC deal 2008; ≈ 3,500 pickers; ≈ 220 t/day recycled; fees ₹65/130/45 (2019); claimed savings ≈ ₹113 crore vs cost ≈ ₹8 crore.	Formalising waste pickers is fairer and cheaper.
9	**Dharavi**	≈ 5,000 units; 50–80% of Mumbai waste, ≈ 60% of plastic (estimates); up to ≈ 250,000 livelihoods; turnover US$650 m–1 bn; earnings ₹300–500/day; threat: redevelopment.	Informal capacity is huge; formalise in situ.
10	**Dell**	2030 goals: 1:1 take-back, 100% packaging, > 50% product content recycled/renewable/low-carbon; packaging 90.2% (FY22); 396.5 m lb sustainable materials (FY22); Goodwill ≈ 620 m lb; self-reported.	Voluntary goal exceeds India's 80% target; audit data.

Must-know terms

EPR: producer is responsible for a product's end-of-life.
PRO: Producer Responsibility Organisation that fulfils EPR for members.
EPR certificate: proof (in kg) that a registered recycler processed waste; bought by producers.
ARF: advance recycling fee paid at sale (Switzerland).
Thermal recovery: burning waste for energy; not material recycling.
Urban mining: recovering metals from e-waste.
Informal sector: unregistered collectors, dismantlers and recyclers.

How to write a 10-mark answer

1. Define the topic and give context (1–2 lines).
2. Strategy: 3–4 key features.
3. Impact: quote 3 numbers with year.
4. SWOT: one point per quadrant.
5. India comparison: 2–3 rows (law, collection, informal sector).
6. Recommendations: 3 practical, India-specific steps.
7. Add a one-line conclusion. Say 'estimated' or 'self-reported' where relevant.